

UNIT 1

Artificial Intelligence

Foundations, History, and Real-World Applications

Introduction to AI · Undergraduate Computer Science / Data Science



What We'll Cover

Three building blocks take you from “what is AI” to “where is AI already working around you.”

01



Introduction to AI

What AI is, how it differs from traditional programming and ML/DL, how an AI system works, and the types of AI.

02



Foundations & History

The disciplines AI draws on, and the milestones — from Turing to Dartmouth to transformers — that shaped the field.

03



Applications of AI

How AI solves real problems across healthcare, finance, transport, NLP, computer vision, generative AI, and more.



01

Introduction to Artificial Intelligence

Defining AI, how it works, and the different types of intelligent systems.

What is Artificial Intelligence?

Simple Definition

Artificial Intelligence is the field of building computer systems that can perform tasks which normally require human intelligence — such as learning, reasoning, perceiving, and decision-making.

Intuition

Instead of telling a computer every exact step, we show it many examples and let it figure out the pattern — much like how a child learns to recognize a dog after seeing many pictures of dogs, not by memorizing a rulebook of “dog-ness.”



Key Technical Point

AI is a broad umbrella term — it covers everything from simple rule-based systems to today's large neural networks. Not all AI involves “learning.”



Analogy

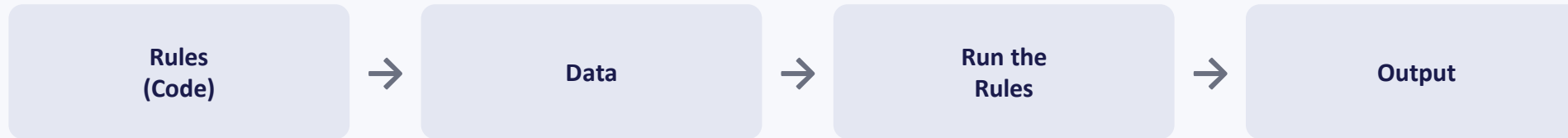
A new employee learning a job:

- Traditional program = following a strict manual, step by step.
- AI system = shadowing experienced staff, noticing patterns, and improving with feedback over time.
- The more relevant experience (data) it sees, the better its judgment (predictions) becomes.

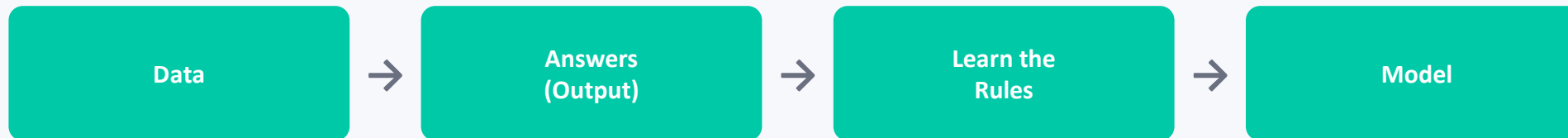
A CORE DISTINCTION

AI vs. Traditional Programming

TRADITIONAL PROGRAMMING



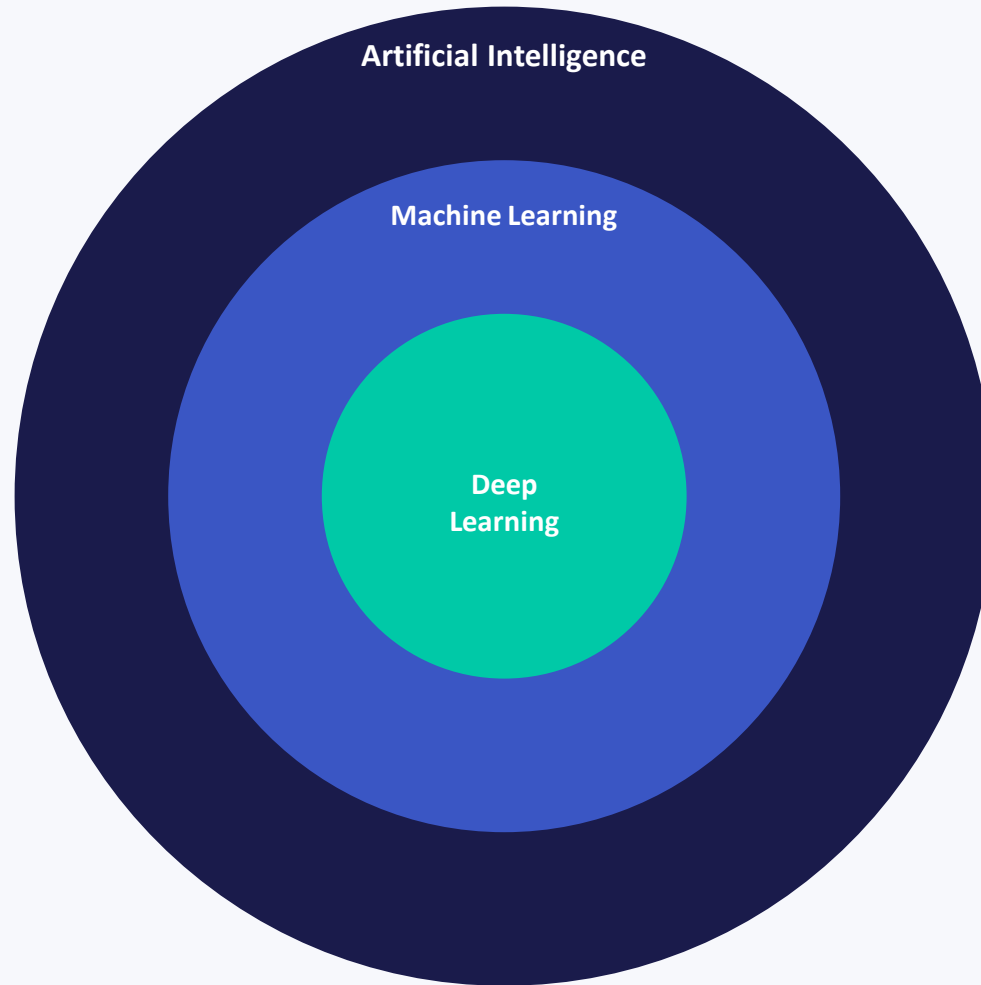
MACHINE / AI-DRIVEN PROGRAMMING



The flip:

In traditional programming, a developer writes explicit rules that transform data into output. In AI/ML, we instead give the system data and known answers, and it learns the rules (the model) itself — which it can then apply to brand-new data it has never seen.

AI vs. Machine Learning vs. Deep Learning



Artificial Intelligence (AI)

The broadest field: any technique that lets machines act intelligently — includes rule-based logic, search, and learning systems.



Machine Learning (ML)

A subset of AI where systems learn patterns from data instead of following hand-written rules (e.g., spam filters, fraud detection).



Deep Learning (DL)

A subset of ML using multi-layered neural networks, powerful for complex data like images, speech, and language.

Comparing AI, ML, and Deep Learning

Aspect	Artificial Intelligence	Machine Learning	Deep Learning
Scope	Broadest field — any intelligent behavior	Subset of AI — learns from data	Subset of ML — layered neural nets
Approach	Rules, search, learning, or hybrid	Statistical pattern learning	Multi-layer neural networks
Data need	Varies (can be little to none)	Moderate to large datasets	Very large datasets
Human input	Often hand-crafted logic	Feature selection by engineers	Learns features automatically
Example	Chess-playing rule engine	Email spam classifier	Face recognition, ChatGPT

Characteristics of AI Systems

Regardless of type, most AI systems share some combination of these capabilities:



Learning

Improving performance from data or experience over time.



Reasoning

Drawing logical conclusions and inferences from available information.



Perception

Interpreting sensory input — images, sound, text, sensor data.



Problem-Solving

Finding solutions or optimal actions in complex situations.



Adaptability

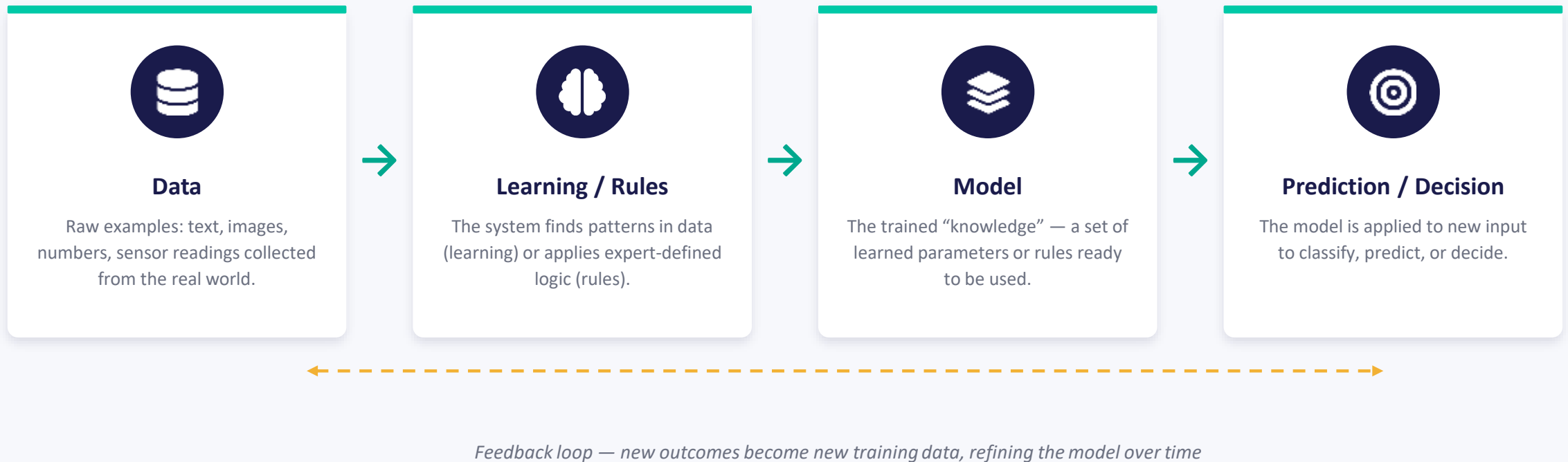
Adjusting behavior as new data or conditions appear.



Autonomy

Acting or deciding with limited or no human intervention.

How an AI System Works



Key point: “Learning” systems build the model from data; “rule-based” systems have the model hand-coded by experts — both still follow this same pipeline.

Types of AI: Narrow, General, and Super

	Narrow / Weak AI	General AI (AGI)	Super AI (ASI)
Definition	Designed for one specific task	Human-level intelligence across any task	Surpasses human intelligence in all domains
Status today	Exists — this is all current AI	Theoretical / research goal	Hypothetical / future concept
Flexibility	Cannot generalize beyond its task	Can transfer learning across domains	Would self-improve beyond human ability
Examples	Siri, Google Translate, ChatGPT, self-driving cars	None yet exist	None yet exist — explored in research & fiction

Nearly everything called “AI” today — including large language models — is Narrow AI: extremely capable at trained tasks, but without general understanding.

Levels of AI: From Reactive to Self-Aware

A separate way to classify AI — by how it uses memory and models of the world:



1. Reactive Machines

React to current input only, no memory of the past. E.g., IBM's Deep Blue chess computer.

IN USE TODAY



2. Limited Memory

Use recent past data to inform decisions. Most AI today, e.g., self-driving cars tracking nearby vehicles.

IN USE TODAY



3. Theory of Mind

Would understand emotions, beliefs, and intentions of others. Still an active research goal.

FUTURE CONCEPT



4. Self-Aware

Would possess consciousness and self-understanding. Purely hypothetical today.

FUTURE CONCEPT

Everyday Examples of AI



Chatbots & Assistants

Siri, Alexa, and customer-support bots understand and respond to natural language.



Recommendations

Netflix, Spotify, and Amazon suggest content based on your past behavior.



Navigation

Google Maps predicts traffic and finds the fastest route in real time.



Face Recognition

Phone unlock and photo-tagging identify faces from images.



Voice Assistants

Speech recognition converts spoken language into text and action.



Spam & Fraud Filters

Email and banking apps flag suspicious messages and transactions.



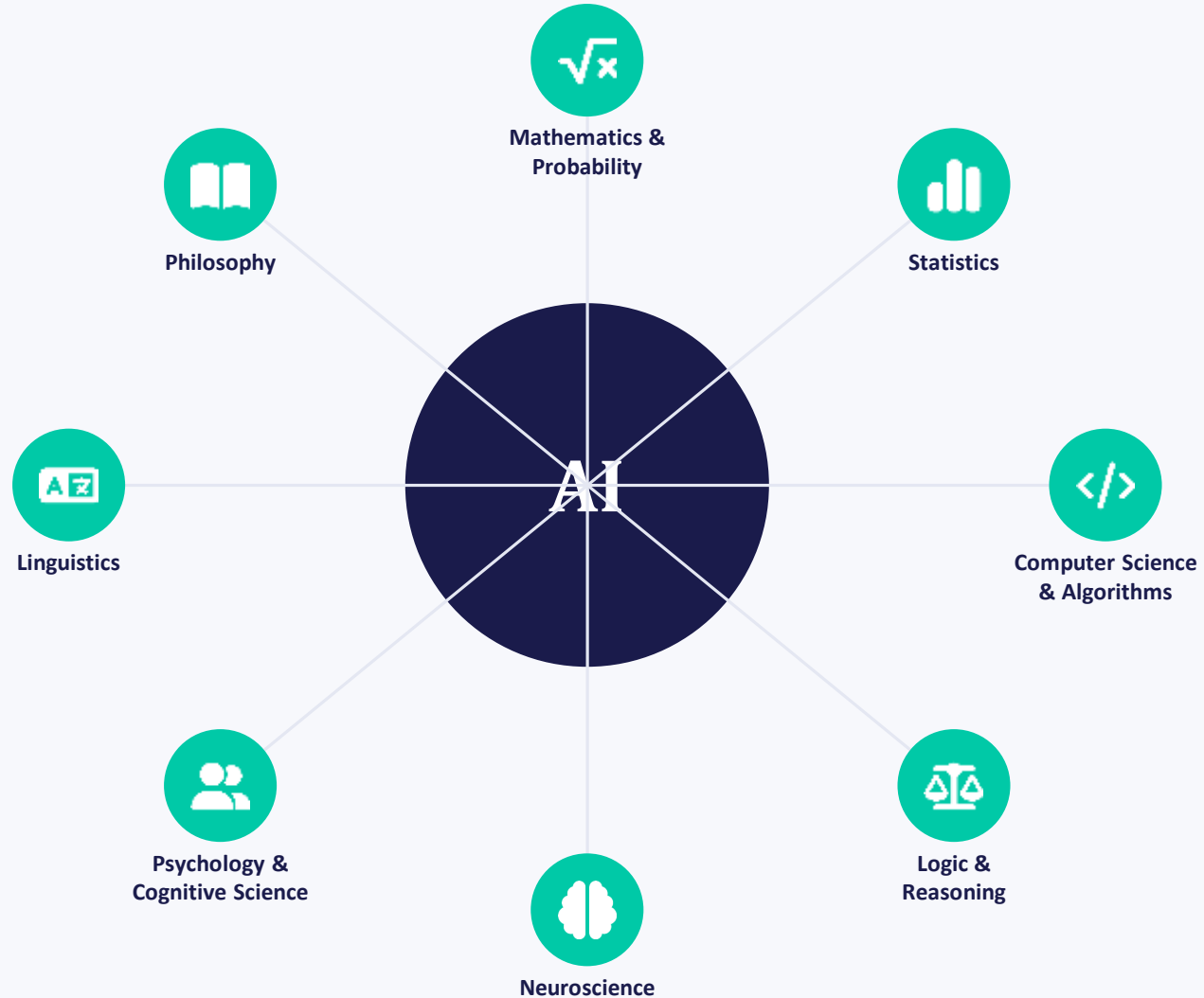
02

Foundations and History of AI

The disciplines AI draws on, and the milestones that built the field.

The Foundations of AI

AI is not one invention — it's the convergence of ideas from many disciplines.



What Each Foundation Contributes



Math & Probability

Linear algebra & calculus power model training; probability handles uncertainty.



Statistics

Provides tools to find patterns, test hypotheses, and evaluate model accuracy.



CS & Algorithms

Supplies data structures, search, and efficient computation that make AI feasible.



Logic & Reasoning

Underpins rule-based systems and formal reasoning about knowledge.



Neuroscience

Inspired neural network design by modeling how brain neurons connect and fire.



Psychology & Cognitive Science

Informs how AI models human learning, memory, and decision-making.



Linguistics

Provides grammar and language structure models behind NLP systems.

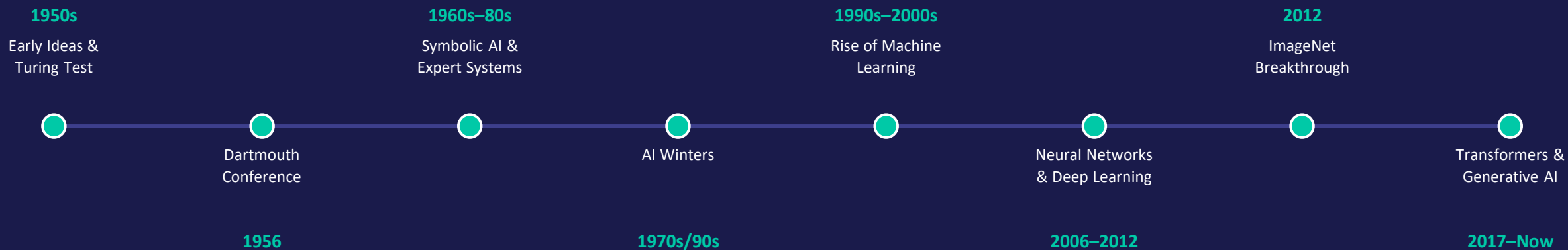


Philosophy

Raises foundational questions of mind, knowledge, and ethics that still guide AI.

A Timeline of Artificial Intelligence

From ancient ideas of thinking machines to today's generative AI — a field shaped by breakthroughs and setbacks.



We'll walk through each milestone next — what happened, why it mattered, and its impact on AI.

Early Ideas & the Turing Test



What Happened

Warren McCulloch & Walter Pitts modeled the first artificial neuron (1943). In 1950, Alan Turing published “Computing Machinery and Intelligence,” proposing a test: if a human judge cannot reliably tell a machine's conversation from a human's, the machine can be said to “think.”



Why It Mattered

It reframed the philosophical question “Can machines think?” into a practical, testable one — shifting AI from speculation toward science, and giving the field its first benchmark for machine intelligence.



Impact on AI

The Turing Test remains a foundational reference point in AI discussions today, and Turing is widely regarded as the father of theoretical computer science and AI.

The Dartmouth Conference



What Happened

John McCarthy, Marvin Minsky, Nathaniel Rochester, and Claude Shannon organized a summer workshop at Dartmouth College. McCarthy coined the term “Artificial Intelligence” for the proposal, which claimed intelligence could, in principle, be precisely described and simulated by a machine.



Why It Mattered

This was the moment AI became a named, organized field of study rather than scattered ideas across mathematics and philosophy. It set an optimistic research agenda that shaped decades of funding and work.



Impact on AI

Widely considered the official “birth of AI” as an academic discipline — the starting point for symbolic AI research over the next two decades.

Symbolic AI, Expert Systems & AI Winters



What Happened

Early AI used symbolic logic — representing knowledge as rules and symbols. This led to Expert Systems (like MYCIN and DENDRAL) that encoded human expert knowledge into “if-then” rules. But when systems failed to scale or meet inflated promises, funding collapsed — periods known as AI Winters (mid-1970s and late-1980s).



Why It Mattered

Expert systems proved AI could deliver real business value in narrow domains. The winters taught the field a hard lesson: overpromising without matching capability damages trust and funding.



Impact on AI

Symbolic techniques still underlie rule-based and reasoning systems today; the winters pushed researchers toward more data-driven, statistical approaches.

The Rise of Machine Learning & Neural Networks



What Happened

Researchers shifted from hand-coded rules toward statistical learning. The backpropagation algorithm (popularized 1986) made training multi-layer neural networks practical. Meanwhile, decision trees, support vector machines, and probabilistic models matured.



Why It Mattered

This marked a philosophical shift: instead of programming intelligence directly, machines could learn it from data. Growing computing power and datasets made this increasingly practical.



Impact on AI

Machine learning became the dominant AI paradigm — the direct ancestor of the deep learning systems used almost everywhere today.

Deep Learning & the ImageNet Breakthrough



What Happened

Geoffrey Hinton and others revived deep neural networks (2006). In 2012, a deep convolutional network (“AlexNet”) won the ImageNet competition by a huge margin, dramatically outperforming traditional computer-vision methods on image classification.



Why It Mattered

It proved that deep learning, combined with GPUs and large labeled datasets, could outperform decades of hand-engineered techniques — triggering an explosion of research investment worldwide.



Impact on AI

Sparked the modern computer-vision revolution (face recognition, medical imaging, self-driving perception) and re-established neural networks as the leading AI approach.

Transformers, Generative AI & the Current Era



What Happened

The 2017 paper “Attention Is All You Need” introduced the Transformer architecture. It enabled large language models (GPT, BERT) and later generative systems that create text, images, audio, and code — including tools like ChatGPT (2022).



Why It Mattered

Transformers scale extremely well with data and compute, allowing a single architecture to power translation, conversation, coding, and image generation — unifying many AI subfields.



Impact on AI

We are now in the generative AI era: AI systems that don't just classify or predict, but create original content, reshaping how people work, learn, and build software.

03

Applications of Artificial Intelligence

Where AI is already solving real problems, across every industry.



The AI Application Map

AI now touches nearly every industry. Here's a snapshot of where it's making the biggest impact.



Healthcare



Finance & Banking



Education



Transportation



E-commerce



Manufacturing



Agriculture



Cybersecurity



NLP



Computer Vision



Generative AI



Entertainment



Smart Homes

Healthcare & Finance



Healthcare

PROBLEM

Diseases are often diagnosed late; analyzing scans and records is slow and expert-limited.

HOW AI HELPS

AI analyzes medical images, patient data, and genetics to detect patterns invisible to the naked eye.

REAL-WORLD EXAMPLE

AI models detect tumors in X-rays/MRIs and predict disease risk from patient history.

BENEFIT

Faster, earlier, more accurate diagnoses — potentially saving lives.



Finance & Banking

PROBLEM

Fraud is hard to catch in real time among millions of daily transactions.

HOW AI HELPS

AI models learn normal spending patterns and flag anomalies instantly.

REAL-WORLD EXAMPLE

Credit card companies use AI to block suspicious transactions within seconds.

BENEFIT

Reduced fraud losses and safer transactions for customers.

Education & Transportation



Education

PROBLEM

Every student learns differently, but classrooms deliver a one-size-fits-all pace.

HOW AI HELPS

AI tracks student performance and adapts content difficulty and pacing to each learner.

REAL-WORLD EXAMPLE

Adaptive learning platforms (e.g., Duolingo, Khan Academy) personalize lesson paths.

BENEFIT

More effective, personalized learning outcomes at scale.



Transportation & Autonomous Vehicles

PROBLEM

Human error causes the majority of traffic accidents; traffic wastes time and fuel.

HOW AI HELPS

AI processes camera/sensor data in real time to perceive, predict, and navigate.

REAL-WORLD EXAMPLE

Self-driving systems (Tesla Autopilot, Waymo) and AI-optimized traffic signals.

BENEFIT

Potential for fewer accidents and more efficient travel.

E-commerce & Manufacturing



E-commerce & Recommendations

PROBLEM

Shoppers are overwhelmed by choice and often can't find what they'd actually like.

HOW AI HELPS

AI analyzes browsing and purchase history to recommend relevant products.

REAL-WORLD EXAMPLE

Amazon's "customers also bought," Netflix's personalized homepage.

BENEFIT

Higher customer satisfaction and increased sales.



Manufacturing & Robotics

PROBLEM

Equipment failures and defects are costly and often caught too late.

HOW AI HELPS

AI predicts maintenance needs and inspects products visually for defects.

REAL-WORLD EXAMPLE

Predictive maintenance sensors and robotic quality-control arms on assembly lines.

BENEFIT

Less downtime, lower costs, and higher product quality.

Agriculture & Cybersecurity



Agriculture

PROBLEM

Farmers must manage crops across large areas with limited visibility into plant health.

HOW AI HELPS

AI-powered drones and sensors monitor soil, crop health, and predict yields.

REAL-WORLD EXAMPLE

Precision farming tools detect pest outbreaks and optimize irrigation.

BENEFIT

Higher yields with less water, pesticide, and labor waste.



Cybersecurity

PROBLEM

Cyberattacks evolve constantly and are too fast and numerous for manual monitoring.

HOW AI HELPS

AI detects unusual network behavior and predicts emerging threats in real time.

REAL-WORLD EXAMPLE

AI-driven intrusion detection systems and automated phishing-email filters.

BENEFIT

Faster threat detection and stronger digital defenses.

Natural Language Processing & Computer Vision



Natural Language Processing (NLP)

PROBLEM

Computers naturally understand numbers, not human language with all its ambiguity.

HOW AI HELPS

AI models learn grammar, meaning, and context to read, translate, and generate text.

REAL-WORLD EXAMPLE

Chatbots, Google Translate, sentiment analysis of customer reviews.

BENEFIT

Machines can understand and communicate in natural human language.



Computer Vision

PROBLEM

Extracting meaning from images and video is trivial for humans but hard for machines.

HOW AI HELPS

AI models learn to detect, classify, and track objects within visual data.

REAL-WORLD EXAMPLE

Face unlock, medical image analysis, quality inspection cameras.

BENEFIT

Machines can “see” and interpret the visual world accurately.

Generative AI, Entertainment & Smart Homes



Generative AI

PROBLEM

Creating original content (text, images, code) traditionally requires human effort.

HOW AI HELPS

Models trained on massive datasets generate new, original content on demand.

EXAMPLE

ChatGPT for writing, DALL-E/Midjourney for images, GitHub Copilot for code.

BENEFIT

Faster content creation and new creative possibilities.



Entertainment & Media

PROBLEM

Studios must predict what audiences want among endless content choices.

HOW AI HELPS

AI analyzes viewing patterns and even assists in content/script generation.

EXAMPLE

Streaming recommendation engines; AI-assisted video editing and special effects.

BENEFIT

More engaging, personalized entertainment experiences.



Smart Homes & Virtual Assistants

PROBLEM

Managing home devices manually is inconvenient and inefficient.

HOW AI HELPS

AI assistants understand voice commands and learn household routines.

EXAMPLE

Alexa, Google Home adjusting lights, temperature, and security automatically.

BENEFIT

Greater convenience, comfort, and energy efficiency.

Key Takeaways

1

AI is the science of building systems that perform tasks requiring human-like intelligence — learning, reasoning, perceiving, and deciding.

5

AI draws on math, statistics, CS, logic, neuroscience, psychology, linguistics, and philosophy — it's inherently interdisciplinary.

2

AI, Machine Learning, and Deep Learning are nested fields: $DL \subset ML \subset AI$, each narrower and more data-hungry than the last.

6

The field evolved through cycles: symbolic AI, AI winters, the rise of machine learning, deep learning, and now generative AI.

3

The core AI pipeline is Data → Learning/Rules → Model → Prediction/Decision, often refined by a feedback loop.

7

AI already delivers real value across healthcare, finance, education, transportation, agriculture, security, and more.

4

All AI in use today is Narrow AI — highly capable at specific tasks, but without general human-like understanding.

Conceptual Questions for Students

Q1

How does an AI-driven approach to solving a problem differ from a traditionally programmed approach? Give an example.

Q2

Why is Deep Learning described as a subset of Machine Learning, which is itself a subset of AI?

Q3

Why is nearly every AI system in use today classified as Narrow AI rather than General AI?

Q4

How did the Dartmouth Conference (1956) change the trajectory of AI research?

Q5

What caused the AI Winters, and what lesson does that history offer about hype in technology?

Q6

Why did the ImageNet breakthrough (2012) matter so much for the future of computer vision?

Q7

In what way do Transformers and generative AI represent a shift from earlier, narrower AI systems?

Q8

Pick one industry from this unit — what specific problem does AI solve there, and what risk might come with it?

Multiple Choice Questions (1–3)

Q1

1. Which of the following best describes Machine Learning's relationship to AI?

ML is broader than AI

✓ ML is a subset of AI

ML and AI are unrelated

ML replaced AI in the 1990s

Answer: ML is a subset of AI — ML is one approach within the broader field of AI — systems that learn patterns from data.

Q2

2. What historic event is widely considered the official birth of AI as a field?

Publication of the Turing Test (1950)

✓ The Dartmouth Conference (1956)

The release of ChatGPT (2022)

The ImageNet competition (2012)

Answer: The Dartmouth Conference (1956) — The 1956 Dartmouth Conference coined the term “Artificial Intelligence” and launched it as an organized field.

Q3

3. All AI systems in active use today (e.g., ChatGPT, Siri) are examples of:

General AI (AGI)

Super AI (ASI)

✓ Narrow / Weak AI

Theory-of-Mind AI

Answer: Narrow / Weak AI — Every current AI system, however capable, is Narrow AI — trained for specific tasks, not general understanding.

Multiple Choice Questions (4–5)

Q4

4. What caused the “AI Winters” of the 1970s and late 1980s?

A lack of computer scientists

✓ Overpromised results not matching real capability, leading to funding cuts

A ban on AI research

The invention of the internet

Answer: Overpromised results not matching real capability, leading to funding cuts — Inflated expectations from symbolic AI and expert systems went unmet, causing funding and interest to collapse.

Q5

5. Which architecture introduced in 2017 enabled modern large language models like GPT and generative AI?

Decision Trees

Convolutional Neural Networks

✓ The Transformer

Expert Systems

Answer: The Transformer — The Transformer architecture (“Attention Is All You Need,” 2017) underlies today's large language and generative models.



Nice work!

Review any missed questions before moving to Unit 2 — where we go deeper into Machine Learning algorithms.

Classroom Case Study & Discussion Activity



Case Study: AI in Hospital Triage

A busy hospital emergency room adopts an AI system that analyzes patient vitals, symptoms, and history to predict which patients need urgent care first — before a doctor examines them.

- The system was trained on 10 years of past patient records from the same hospital.
- It flags patients as Low, Medium, or High urgency within seconds of intake.
- Nurses report the tool is usually right — but occasionally misses urgent cases in patients whose symptoms are rare in the training data.



Discuss in Groups

- Which foundation(s) from Section 2 (e.g., statistics, logic, psychology) does this system rely on most?
- Is this Narrow AI, and how can you tell?
- Why might it underperform on “rare” symptoms? Connect this to how AI systems learn from data.
- What safeguards would you recommend before fully trusting this system's rankings?